

Java Arrays & Patterns Tasks

Complete all 6 programs in Java.

1. ARRAYS

1. Find the Largest Element in an Array

Scenario: A teacher wants to find the highest mark scored by a student from a list of marks.

Input:

```
Enter number of elements: 5
Enter elements: 45 78 32 90 67
```

Output:

```
Largest element = 90
```

2. Find the Sum of Array Elements

Scenario: A shopkeeper has the prices of 5 products and wants to calculate their total price.

Input:

```
Enter number of elements: 5
Enter elements: 10 20 30 40 50
```

Output:

```
Sum = 150
```

3. Count Even and Odd Numbers

Scenario: A teacher wants to separate the number of even and odd numbers from a list of numbers.

Input:

```
Enter number of elements: 6
Enter elements: 12 7 8 15 20 9
```

Output:

```
Even = 3
Odd = 3
```

4. Reverse an Array

Scenario: A system stores numbers in an array and needs to display them in reverse order.

Input:

```
Enter number of elements: 5
Enter elements: 10 20 30 40 50
```

Output:

```
50 40 30 20 10
```

2. PATTERNS

5. Hollow (Halo) Square Pattern

Scenario: Design a square-shaped pattern where only the boundary contains stars and the inside remains empty.

Input:

Enter size: 5

Output:

```
* * * * *
*      *
*      *
*      *
*      *
* * * * *
```

6. Slant Star Pattern

Scenario: Create a right-slanting pattern using stars, where the stars move one position to the right on each new line.

Input:

Enter number of rows: 5

Output:

```
*
 *
  *
   *
    *
```

Instructions:

- Complete all 6 programs using Java.
- Use proper class names and meaningful variable names.
- Test each program with the given sample input and output.
- Push the completed code to the assigned GitHub branch.